

PRODUCT REQUIREMENTS DOCUMENT (PRD)

Disaster Relief Coordination Platform

Government of India

Live URL: <https://disasterhelper.dpdns.org>

1. Overview

The Disaster Relief Coordination Platform is a government-backed web application designed to coordinate disaster relief operations across India. The platform enables volunteers, NGOs, and administrators to collaborate in real-time for managing relief requests, resources, incidents, SOS alerts, and volunteer schedules.

The system provides geospatial tracking, real-time communication, emergency broadcasting, resource allocation, and disaster zone monitoring to ensure effective disaster response and relief management.

2. Technology Stack

Layer	Technology
Frontend	React 18, Vite, React Router v6, Leaflet.js, i18next, Socket.IO Client
Backend	Express.js, Node.js, Socket.IO, Mongoose ODM, JWT, Joi, Multer, Helmet
Database	MongoDB Atlas with 2dsphere Geospatial Indexes
PWA	Installable Progressive Web Application with Offline Support
Deployment	Render.com

3. User Roles

3.1 Volunteer

- Create Relief Requests
- Manage Own Requests
- Claim Open Requests
- Participate in Chat
- Upload Files
- Submit Feedback
- View Assigned Schedule

3.2 NGO

- All Volunteer Features
- Resource Management
- Relief Coordination
- Request Fulfillment

3.3 Administrator

- Full System Access
 - User Management
 - Zone Management
 - Incident Management
 - SOS Management
 - Escalation Handling
 - CSV Import/Export
 - Statistics Dashboard
 - Platform Monitoring
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4. Functional Requirements

4.1 Authentication & Authorization

Features

- JWT-based Authentication
 - Token Expiry: 7 Days
 - Password Encryption using bcrypt (12 rounds)
 - User Registration (Volunteer/NGO)
 - Admin Account Seeded through Environment Variables
 - Role-Based Access Control (RBAC)
 - Protected Client and Server Routes
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4.2 Relief Request Management

Features

- Create Request
- Read Request
- Update Request
- Delete Request
- View Request Details
- Claim Request
- Unclaim Request
- Escalate Request
- Upload Attachments
- Request Comments

Request Fields

- Title
- Description
- Category (14 Categories)
- Priority
 - Critical
 - High
 - Medium
 - Low
- People Count
- Geo Location
- Attachments

Status Workflow

Open → In Progress → Fulfilled

File Upload Rules

- Maximum 5 Files
- Maximum Size: 10 MB per File
- Supported Formats:
 - JPG
 - PNG
 - PDF
 - DOC
 - DOCX

Additional Features

- Interactive Map Picker
 - GPS Location Detection
 - Real-Time Chat
 - Audit Trail Logging
 - Escalation Workflow
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4.3 Resource Management

Features

- Resource Inventory Tracking
- Resource Allocation
- Resource Deallocation
- Resource Search
- Resource Status Monitoring

Resource Fields

- Name
- Category
- Quantity
- Unit
- Location
- Status

Auto Status Logic

Quantity	Status
High	Available
Low	Low Stock
0	Depleted

Smart Matching Algorithm

Score Formula:

- Category Match = 60%
- Distance Match = 40%

Conditions:

- 100 KM Search Radius
 - Haversine Distance Calculation
 - Top 10 Results Returned
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4.4 Disaster Zone Management

Features

- Create Zone
- Update Zone
- Delete Zone
- View Zone
- Coverage Analysis
- Heat Map Visualization

Zone Properties

- Center Coordinates
- Radius (1–500 KM)
- Severity
 - Critical
 - High
 - Medium
 - Low
- Disaster Type
- Affected Population

Coverage Status

- Covered
 - Partial Coverage
 - Resource Gap
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4.5 Incident Management

Features

- Create Incident
- Update Incident
- Delete Incident
- View Incident
- Incident Statistics

Incident Fields

- Name
 - Disaster Type
 - Severity
 - Status
 - Linked Zones
 - Affected Population
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4.6 SOS Alert System

Features

- One-Click Emergency Broadcast
- Real-Time Notification
- Admin Acknowledgment
- Resolution Workflow

Status Flow

Active → Acknowledged → Resolved

4.7 Real-Time Chat

Features

- Request-Based Chat Rooms
- Socket.IO Communication
- Message History
- Pagination Support

Message Types

- User Message
 - System Message
 - SOS Message
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4.8 Volunteer Scheduling

Features

- Schedule Creation
- Shift Assignment
- Skill Tagging
- Zone Assignment

Shift Types

- Morning
- Afternoon
- Night
- Full Day

Status Flow

Scheduled → Active → Completed

4.9 Geofencing

Features

- Radius-Based Search
- Nearby Resource Search
- Nearby Request Search
- Nearby Zone Search

Technical Implementation

- MongoDB \$geoWithin
 - \$centerSphere Queries
 - Haversine Distance Formula
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4.10 Feedback System

Features

- Rating Submission
- Comments
- Delivery Confirmation

Rules

- One Feedback per User per Request
 - Rating Scale: 1–5 Stars
 - Delivery Confirmation Auto-Fulfills Request
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4.11 Bulk Import & Export

Features

- CSV Export
- CSV Import
- Template Download
- Data Preview

Import Features

- Delimiter Detection
 - Column Alias Mapping
 - Row Validation
 - Editable Preview Table
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4.12 Admin Dashboard

Dashboard Modules

Statistics

- Total Users
- Total Requests
- Total Resources
- Total Incidents
- Daily Trends
- Bar Charts

User Management

- Change Roles
- Delete Users
- Search Users

Request Management

- Filter Requests
 - Escalation Monitoring
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4.13 Internationalization (i18n)

Supported Languages:

1. English
2. Hindi
3. Bengali
4. Tamil
5. Telugu
6. Marathi
7. Gujarati
8. Kannada
9. Punjabi

Features:

- Language Persistence
 - Dynamic Translation Loading
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4.14 Notification System

Features

- WebSocket Notifications
- Notification Bell
- Unread Badge
- Mark All Read
- Clear All Notifications

Storage

- Local Storage Persistence
 - Maximum 50 Notifications
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5. API Summary

Module	Endpoints	Authentication
Auth	4	Mixed
Requests	10	Required
Resources	8	Required
Zones	6	Mixed
Incidents	5	Mixed
SOS	4	Required
Chat	3	Required
Feedback	3	Required
Schedules	4	Required
Escalation	3	Admin
Geofencing	1	Required
Bulk Operations	4	Admin
Admin	6	Admin

Total API Endpoints: 59

6. Data Models

User

- email
- passwordHash
- role
- displayName
- skills[]
- phone
- notifications[]

Request

- title
- description
- location (GeoJSON)
- status
- category
- priority
- peopleCount
- files[]
- comments[]
- auditLog[]
- escalationFields

Resource

- name
- category
- quantity
- unit
- location (GeoJSON)
- status
- allocations[]

Zone

- name
- center (GeoJSON)
- radiusKm
- severity
- status
- disasterType
- affectedPopulation

Incident

- name
- disasterType
- severity
- status
- zones[]
- affectedPopulation

SosAlert

- message
- location
- userId
- status

ChatMessage

- requestId
- sender
- text
- type

Feedback

- requestId
- rating
- comment
- deliveryConfirmed

Schedule

- userId
 - zoneId
 - startDate
 - endDate
 - shift
 - skills[]
 - status
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7. Security Requirements

Security Controls

- Helmet Security Headers
- CSP Policy
- CORS Whitelisting
- JWT Authentication
- Socket.IO JWT Validation
- Rate Limiting

Rate Limits

Authentication APIs:

- 20 Requests per 15 Minutes

Additional Protections

- Maximum Request Body Size: 10 MB
 - Admin Cannot Delete Own Account
 - One Feedback per User per Request
 - Input Validation using Joi
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8. User Interface & Design

Theme

Government of India Inspired Design

Color Palette

Color	Hex Code
Navy Blue	#000080
Saffron	#FF9933
Green	#138808
White	#FFFFFF

UI Requirements

- Fully Responsive Design
- Mobile Hamburger Navigation
- Interactive Maps
- Real-Time Notifications
- Accessible Components
- Progressive Web App Support

Footer Information

Emergency Helplines:

- Emergency Response: 112
 - Ambulance Services: 108
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9. Success Metrics

- Faster Disaster Response Time
- Increased Volunteer Participation
- Efficient Resource Allocation
- Reduced Relief Request Resolution Time
- Improved Coordination Between NGOs and Government Agencies
- Real-Time Visibility of Disaster Operations

10. Conclusion

The Disaster Relief Coordination Platform provides a comprehensive solution for managing disaster response operations across India. By integrating geospatial technologies, real-time communication, intelligent resource matching, volunteer scheduling, and emergency broadcasting, the platform enables efficient coordination between citizens, NGOs, and government authorities during disasters.